DEEPAK KATUKURI

AI SOFTWARE ENGINEER · FULL-STACK · APPLIED AI / LLM



CONTACT

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TECHNICAL SKILLS

Languages: Python, JavaScript, SQL, PL/SQL, Java, C++

Frontend: React, Vite, Tailwind CSS, React Query, Recharts

Backend & APIs: FastAPI, REST APIs, PostgreSQL, Docker, Kubernetes

AI / LLM: LLM API integration (Ollama; OpenAI / Anthropic-compatible patterns), prompt engineering, RAG, embeddings, vector search, CLIP, DistilBERT

AI-Assisted Dev: coding agents, LLM-accelerated workflows, rapid prototyping

DevOps & Cloud: Git / GitHub, CI/CD, AWS, pytest

ML / Deep Learning: PyTorch, TensorFlow, scikit-learn, XGBoost, CNNs, fine-tuning, NumPy, Pandas

EDUCATION

MSc Data Science
University of Europe, Germany
Sep 2025 – Present

BTech – CS & Engineering
MLRIT, India
Jun 2018 – Jul 2022

CERTIFICATIONS

PL/SQL – Database Management
Core Java – Software Development
OneShield – P&C Insurance

LANGUAGES

English – Fluent
German – Beginner (A1/A2)

PROFILE

AI Software Engineer and Data Science MSc candidate who ships full-stack products fast. I build applications end-to-end — React / Vite frontends on FastAPI + PostgreSQL backends — and use AI-assisted development (LLM APIs, prompt engineering, coding agents) to move quickly from concept to deployment. A framework-free builder with strong programming fundamentals, comfortable in any modern language and backed by 2.5 years of production software engineering. Seeking a working-student AI Software Engineer role in Berlin, owning projects end-to-end.

PROJECTS

RA Nutrition & Lifestyle App — Full-Stack Health Product

React · Vite · Tailwind CSS · React Query · Recharts · FastAPI · PostgreSQL

- Architected and built a full-stack health app end-to-end — React / Vite + Tailwind frontend on a FastAPI + PostgreSQL backend — with JWT authentication and token refresh, meal / symptom logging, and a transparent rule-based recommendation engine.
- Designed a normalized data model across 4 logical PostgreSQL schemas (core, static, tracking, intelligence) and shipped a weekly analytics dashboard visualizing meal and symptom trends using React Query data-fetching and Recharts.
- Architected a V3 ML tier (scikit-learn / XGBoost + PyTorch LSTM / GRU / TCN) for flare-risk prediction and pain-score forecasting, gated to ship only where it beats the rule-based baseline.

RAG Application — Retrieval-Augmented Generation from Scratch

Python · Ollama / LLM APIs · sentence-transformers · github.com/Deepak589/RAG_pipeline

- Shipped a framework-free RAG application in ~270 lines of Python (no LangChain or vector DB), integrating local LLM inference via Ollama with the same request / response pattern as OpenAI / Anthropic chat APIs and degrading gracefully when the model was offline.
- Cut repeat embedding cost with a SHA-256-fingerprinted vector cache that auto-invalidates on chunk / model change, pairing a dense bi-encoder retriever (all-MiniLM-L6-v2, cosine top-3) with a TF-IDF baseline to raise relevance.

Food Image-to-Recipe Retrieval — Multimodal AI

PyTorch · CLIP ViT-B/32 · DistilBERT · InfoNCE · MSc final project

- Prototyped and shipped a config-driven multimodal retrieval app end-to-end, reaching a validation median rank of 10/200 on a 13.5K-image dataset via symmetric InfoNCE, and accelerated training ~10x by caching CLIP image features. (0.98 ROC-AUC EfficientNetB0 CT-scan classifier also built — PyTorch / timm.)

EXPERIENCE

Software Engineer — ValueMomentum
Mar 2022 – Sep 2024

- Owned features across 2 enterprise insurance applications from concept to deployment on OneShield PAS, resolving production defects across policy workflows.
- Built and optimized PL/SQL stored procedures and functions powering backend data operations and endorsement modules serving live policy processing.
- Accelerated releases by running QA/UAT test cases and supporting CI/CD deployment of fixes across the full policy lifecycle.

Machine Learning Intern — SmartBridge × IBM
Jun – Jul 2019

- Built and deployed a lung-cancer risk-prediction ensemble (Random Forest, KNN, Gradient Boosting, Neural Network) via IBM Watson ML, communicating results with Matplotlib / Seaborn visualizations.